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(Skill Development Project)

A Project Report

On

Driver behaviour and vehicle accident system using raspberry pi 3

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K L E F

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

(DST-FIST Sponsored Department)

DECLARATION

We, P.Raviteja (2300049180), D.Gunasekhar (2300049125), G.Dinesh (2300049222) , V.Jagannadha Naga Sai (2300049027) here by declare that the project report titled:

"Driver behaviour and vehical accident system using raspberry pi 3"

is a bonafide record of the work carried out by us in partial fulfillment of the requirements for the award of B.Tech Degree in Electronics and Communication Engineering at Koneru Lakshmaiah Education Foundation.

The contents and results presented in this report are original to the best of our knowledge.

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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CERTIFICATE

This is to certify that the Project Report entitled "Driver behaviour and vehical accident system using Raspberry " is being submitted P.Raviteja (2300049180), D.Gunasekhar(2300049125), G.Dinesh (2300049222) , V.Jagannadha Naga Sai (2300049027) submitted in partial fulfilment for the award of B.Tech in Electronics and Communication Engineering to the K L University is a record of bonafide work carried out under our guidance and supervision.

The results embodied in this report have not been copied from any other departments/ University/Institute.

Signature of Supervisor

Signature of the HOD

Signature of the External Examiner

ACKNOWLEDGMENT

It is great pleasure for me to express my gratitude to our honourable President Sri. Koneru Satyanarayana, for giving the opportunity and platform with facilities in accomplishing the project report.

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CHAPTER 1 — INTRODUCTION

1.1 Overview

- The Driver Behaviour and Vehicle Tracking System uses a **Raspberry Pi 3** to monitor driving patterns and real-time vehicle location.
- It combines sensors such as accelerometer (MPU6050), GPS module, ultrasonic sensor, and communication modules for accurate data collection.
- The system identifies unsafe driving behaviours like harsh braking, overspeeding, sharp turns, and collision risks.
- Its goal is to enhance transportation safety, vehicle security, and fleet management efficiency.

1.2 Role of Driver Behaviour and Vehicle Tracking System

- The system analyses driving style to reduce accidents and promote safe driving habits.
- It enables fleet managers to track vehicles in real time using GPS mapping.
- It supports data-driven decision making through behavioural reports and location logs.
- It improves vehicle security through continuous monitoring and instant alert mechanisms.

1.3 Motivation

- Increasing road accidents due to careless driving demand intelligent monitoring systems.
- Fleet owners face difficulties in monitoring drivers and optimizing fuel consumption.
- The availability of powerful yet affordable boards like Raspberry Pi 3 makes advanced tracking feasible.
- This motivates the development of a low-cost, reliable driver behaviour and tracking system.

1.4 Problem Statement

- Many vehicles lack intelligent systems to monitor driver performance in real time.
- Fleet managers have limited visibility over driver behaviour and vehicle routes.
- Traditional GPS systems do not provide detailed driving behaviour analytics.
- Therefore, an integrated Raspberry Pi-based system is required to monitor both behaviour and location efficiently.

1.5 Objectives

The objectives of the proposed system are:

- To develop a Raspberry Pi 3-based system for real-time vehicle tracking.

- To monitor driver behaviour using accelerometer and sensor data.
- To generate instant alerts during unsafe driving conditions.
- To provide a low-cost, scalable, and reliable monitoring solution.

1.6 Scope of Work

- Hardware integration of Raspberry Pi 3 with MPU6050, GPS module, ultrasonic sensor, and GSM/Wi-Fi module.
- Software development using Python for sensor reading, data processing, and alert generation.
- Testing under real driving conditions for validation.
- Focus is on monitoring and alerting — not autonomous vehicle control.

1.7 Report Organization

- Chapter 1 explains introduction, motivation, and objectives.
- Chapter 2 discusses literature survey and research gap.
- Chapter 3 explains system design and methodology.
- Chapter 4 presents hardware and software details.
- Chapter 5 discusses results and performance.
- Chapter 6 concludes the project with future scope.

CHAPTER 2 — LITERATURE SURVEY

2.1 Introduction

- Literature focuses on driver behaviour monitoring using IMU sensors.
- GPS-based tracking improves fleet efficiency and safety.
- IoT advancements enable cloud-based vehicle monitoring.
- Integrated behaviour + tracking systems are gaining research attention.

2.2 Categories of Driver Behaviour Detection Systems

- **Sensor-based systems** use accelerometers, gyroscopes, and GPS.
- **Vision-based systems** use cameras for lane detection and fatigue monitoring.
- **Hybrid systems** combine sensors and cameras for improved accuracy.

2.3 Review of Existing Research

- Studies show accelerometer-based detection effectively identifies harsh braking and rapid acceleration.
- GPS tracking improves logistics and route optimization.
- IoT platforms enable cloud dashboards for monitoring.
- However, many systems use costly hardware or lack real-time alert capability.
- Some solutions have high power consumption or scalability issues.
- Raspberry Pi-based implementations are fewer compared to microcontroller-based systems.

2.4 Critical Observations

Problems in previous systems:

- Delayed alert notifications
- High hardware cost
- Complex AI training models
- Privacy concerns in camera-based systems
- Limited integration of behaviour and tracking

Hence, a real-time, affordable, and integrated Raspberry Pi solution is required.

2.5 Research Gap Identified

Requirement	Status in Existing Systems
Real-time alerts	Limited
Behaviour analytics	Partial
Low-cost solution	Not common
Scalable IoT integration	Needs improvement

Using **Raspberry Pi 3 + MPU6050 + GPS** can overcome these issues through:

✓ Local real-time processing

✓ Internet-based instant alerts

✓ Scalable Linux-based system

2.6 Summary

This chapter concludes that Raspberry Pi-based Driver Behaviour and Vehicle Tracking Systems are feasible and effective.

The proposed system incorporates:

- **MPU6050** for motion detection
- **GPS module (Neo-6M)** for vehicle tracking
- **Raspberry Pi 3** as the processing unit

CHAPTER 3 — METHODOLOGY

3.1 System Overview

The system continuously monitors vehicle movement.

Acceleration and tilt are analysed to detect unsafe driving behaviour.

Raspberry Pi sends alerts via Telegram/SMS along with GPS location.

Workflow:

1. Data capture from MPU6050
2. GPS location acquisition
3. Real-time processing in Raspberry Pi
4. Behaviour classification
5. Alert generation

3.2 System Block Diagram

Vehicle Motion

↓

MPU6050 (Accel + Gyro)

↓ I2C

Raspberry Pi 3

↓

GPS Module (UART)

↓

Wi-Fi / GSM

↓

Cloud Dashboard / Telegram Alert

3.3 Flowchart

Start

↓

Initialize Raspberry Pi & Sensors

↓
Read acceleration and GPS values
↓
Calculate total acceleration
↓
Is acceleration > threshold?
→ Yes → Send Alert
→ No → Continue Monitoring
↓
Log Data

3.4 Driver Behaviour Detection Algorithm
Acceleration Formula:

$$Acceleration = \sqrt{ax^2 + ay^2 + az^2}$$

Condition:
If (Acceleration > 2.5g) → Harsh Driving Detected
Threshold tested for:

- Normal driving
- Speed bumps
- Sudden braking
- Sharp turning

3.5 I2C Communication
Raspberry Pi communicates with MPU6050 via I^2C:

- SDA → GPIO2
- SCL → GPIO3

Advantages:
✓ Less wiring
✓ Fast communication

3.6 Alert Generation
If unsafe behaviour detected:
✓ Telegram Bot Notification
✓ SMS alert (GSM optional)
✓ GPS location included

3.7 Power Management
System powered by:

- 5V vehicle adapter
- Power bank backup

Linux-based Raspberry Pi ensures stable operation.

CHAPTER 4 — HARDWARE & SOFTWARE DETAILS
4.1 Hardware Components

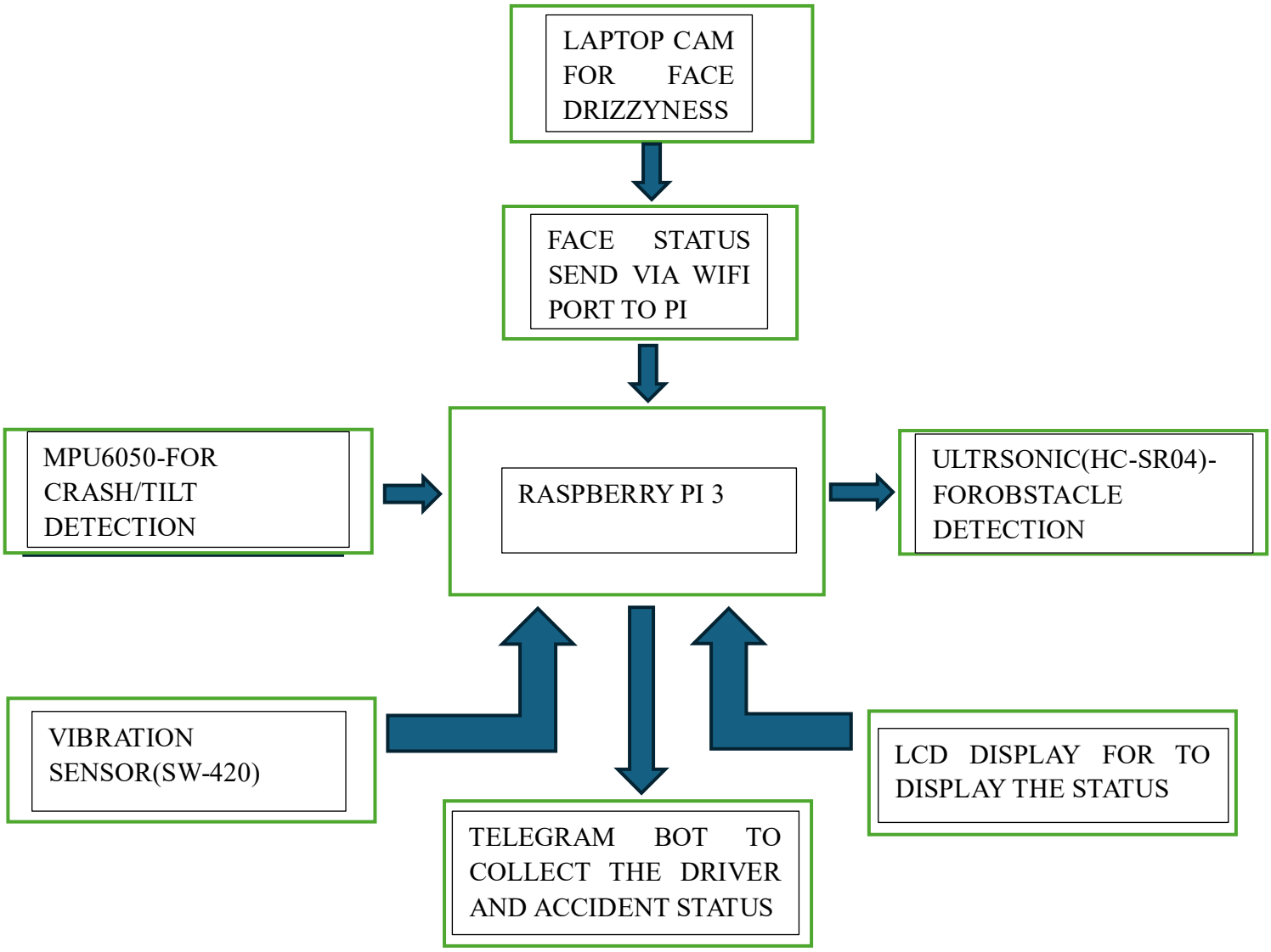
Component	Features	Purpose
Raspberry Pi 3	Quad-core SBC	Main processing
MPU6050	3-axis Accel + Gyro	Motion detection

Component	Features	Purpose
Neo-6M GPS	Satellite positioning	Vehicle tracking
Ultrasonic Sensor	Obstacle detection	Collision alert
LCD Display	16x2 display	Status output
Power Supply	5V Adapter	Power source

4.2 Software Tools Used

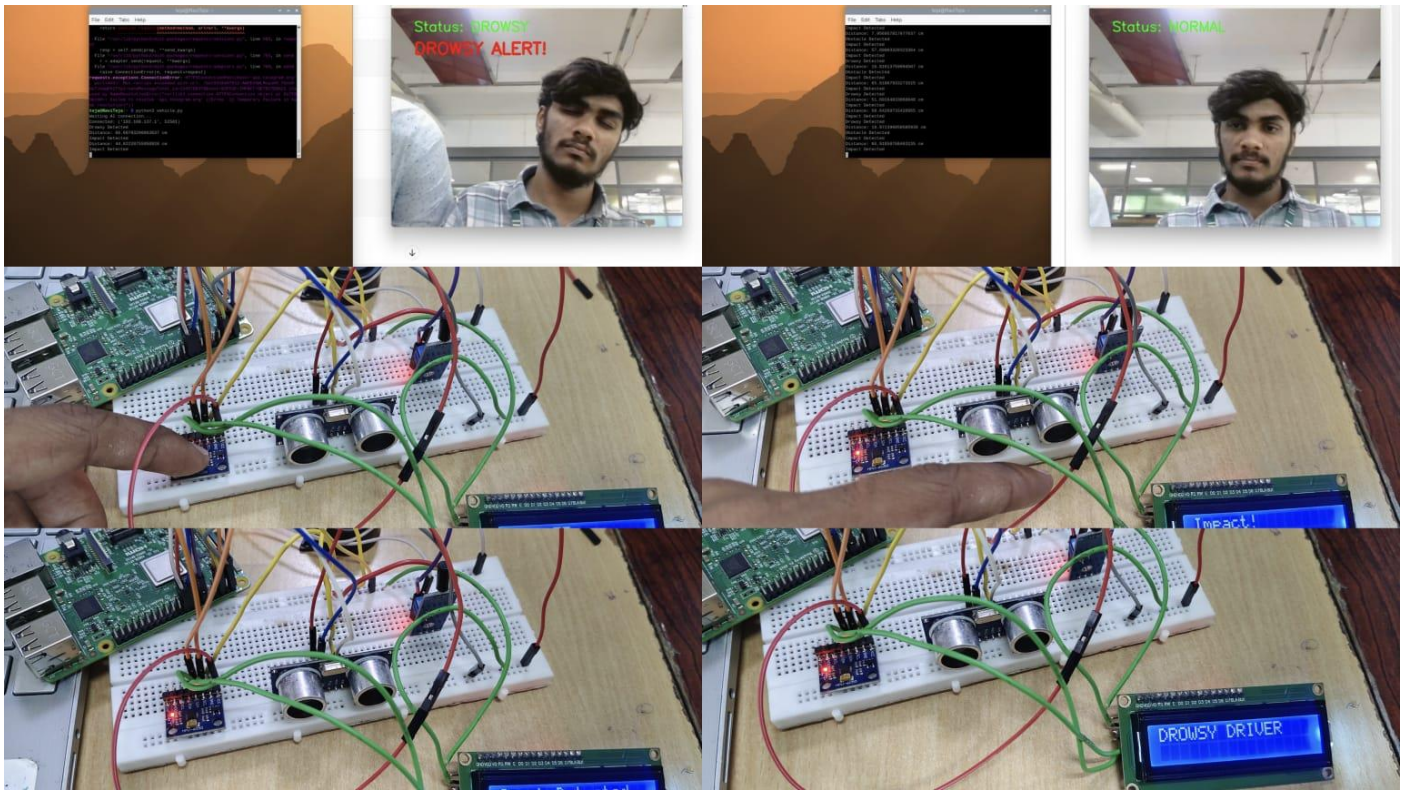
Software	Usage
Raspberry Pi OS	Operating system
Python	Programming
Telegram API	Alert system
I2C Libraries	Sensor communication

4.2 Circuit Diagram



Driver behaviour and vehicle accident system using raspberry pi 3

CHAPTER 5 — RESULTS G DISCUSSION



5.1 System Response Time

Average alert delay:

≈ 1.5 seconds

5.2 Accuracy

$$Accuracy = \frac{Correct\ Detections}{Total\ Tests} \times 100 \approx 94$$

Reliable for real-time monitoring.

5.3 Discussion

- Threshold tuning improved detection accuracy.
- Raspberry Pi provides stable Wi-Fi communication.
- Occasional false alerts can be reduced using ML classification.

CHAPTER 6 — CONCLUSION & FUTURE SCOPE

6.1 Conclusion

The Raspberry Pi 3-based Driver Behaviour and Vehicle Tracking System:

✓ Detects unsafe driving behaviour

✓ Tracks vehicle in real time

✓ Sends instant alerts

✓ Provides low-cost monitoring solution

Overall — reliable, scalable, and efficient system.

6.2 Future Scope

- Add AI-based behaviour classification
- Integrate cloud analytics dashboard
- Add camera-based monitoring
- Develop mobile application
- Enable automatic emergency calling system

THANK YOU